N2U5/5A21

Reg. No									

St. Joseph's College of Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore – 607001

B.Sc DEGREE EXAMINATION APRIL 2021

ASMT202 - MATHEMATICAL STATISTICS - II

Time : 3 hrs **Max Marks :75**

SECTION - A (10x1 = 10)

Answer ALL the questions

1. Define Binomial distribution.

2. Define Geometric distributions.

3. Define Hypothesis.

4. What are the application for 't' test?

5. What is Estimation theory?

6. What is Interval Estimation?

7. What is type I even?

8. Write the formula for calculating standard error.

9. What is confidence interval?

10. What is small and large sample?

SECTION – B (5X5=25)

Answer **ALL** the Questions

11. (a) In a binomial distribution consisting of 5 independent trials of probabilities of 1 and 2 successes are 0.4096 and 0.2048 respectively. Find the parameter 'p' of the distribution.

(Or)

(b) In a poisson distribution, frequency corresponding to 3 successes is 2/3 times frequency corresponding to 4 successes. Find the mean and standard deviation of the distribution. .

12. (a) Obtain mean and variance of exponential distribution .

(Or)

(b) Obtain the relationship between t and f distribution.

13 (a) What are the properties of a good estimator?

(Or)

(b) Explain the methods of maximum likelihood.

14. (a) A random sample of 10 boys had the following I,Q is 61, 64, 65, 62, 66, 65, 60, 64, 68 and 66. Is it reasonable to believe that the average I.Q is greater than 66?

(Or)

(b) Explain the F test for differences of variances.

15. (a) In a sample of 1000 people in Maharashtra, 540 are rice eaters and rest are wheat eaters. Can we assume that both rice and wheat eaters are equally popular in this state at 1% level of significance?

(Or)

(b) Explain the procedure for testing of significance for difference in means for large samples.

SECTION - C (3X10=30)

Answer any **THREE** Questions

16. Derive probability density function of Poisson distribution.

17. Obtain the probability density function of chi-square distribution with n degrees of freedom.

18. State and prove Creamer – Rao Inequality.

19. The marks of students before and after coaching are given below.

Before Coaching	60	64	50	45	40	56	50	47
After Coaching	70	72	60	50	48	62	59	55

Is the coaching effective?

20. On the basis of their total scores, 200 candidates of a civil service examination are divided into two groups. The upper 30% and the remaining 70%. Consider the first question in the examination. Among the first group, 40 had correct answers, whereas among the second group 80 had correct answers. On the basis of these results, can one conclude that first question is no good at discriminating ability of the type being examined here.

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